

GIST: A Gated Insight Synthesis Topology, Native in a Coherence Calculus and Trainable as the Recurrent State-Space/Convolutional Hybrid It Already Is

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github.com/ETEllis/bidi-coherence-delta-calculus (substrate)

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Abstract

GIST (Gated Insight Synthesis Topology) is a synthesis mechanism for evidence-bearing reasoning: claims decompose into a 64-slot lattice ($2^6=4^3$), evidence superposes as phases, a corefold pyramid distills each scope’s triad to a singular conclusion, and a guarded *balanced-ternary commit* quantizes it—+ asserted, − opposed, 0 a live open aperture—behind a barrier that holds contradiction-debt rather than latching it. We present GIST built *entirely* out of the BiDi Coherence-Delta Calculus (CDC): the shipped mechanism is native `.cdc` source executed by CDC’s existing C micro-bridge with zero host code of its own (machine-checked `host-debt` 0; 83/83 contract expectations; 70 native witnesses). Pearl’s do-operator requires no new primitive: an intervention is a latched cell plus a severed inbound channel, and we exhibit necessity, sufficiency-restoration, backdoor severance, and frontdoor mediation as runtime-checked twin-field programs. We then observe that the mechanism’s forward dynamics already constitute a neural architecture—a selective nonlinear state-space recurrence (Kuramoto phase turn) over $SO(2)$ -weighted graph convolutions with a ternary bottleneck and hypercube pooling—and train that hybrid (109 parameters) against labels produced by the native runtime itself. Initialized *at the calculus*, it reaches 0.984 slot accuracy and 0.868 global-verdict accuracy on held-out 64-slot scenes (majority baselines 0.661/0.532), and the trained weights export back to `.cdc`, where the same C runtime reproduces the model’s conclusions on all 27 ternary inputs (27/27 parity). Learning itself is shown native: the runtime now executes CDC’s declared Hebbian plasticity as part of flow.

1 Introduction: GIST before any neural network

GIST is, at its conceptual base, a discipline for concluding under evidence: (i) *scope* every question into a finite lattice of micro-scopes; (ii) *superpose* evidence within each scope so that agreement reinforces and contradiction cancels; (iii) *distill* each scope’s evidence triad through a fixed pyramid ($3 \rightarrow 2 \rightarrow 1$) into a singular conclusion; (iv) *commit* conclusions only through guards—a nonnegative-balance barrier over the committed word (leading contradiction is debt and *holds* with attribution rather than latching) and an energy guard (a commit may not increase the scope’s variational potential); (v) *gate* any release of a synthesis on coverage, parity, stability, counterfactual survival, causal lift, global agreement, and retrieval saturation; and (vi) *reduce* the lattice through a Hamming ladder whose reduction tree doubles as a Merkle verification tree, ending in a council-reconciled, stamped decision. The balanced-ternary middle value is

load-bearing: 0 is a live crossing—“still open, contribute evidence”—not falsehood, giving the mechanism a first-class state that language models notoriously lack.

This paper reports GIST’s construction *from first principles* inside an existing formal substrate, and the consequence that follows: the mechanism is already a neural network, and can be trained as one without leaving its own semantics.

2 The substrate: BiDi Coherence-Delta Calculus

CDC [1] is a native-language/kernel pair whose carrier is a coherence algebra $\mathcal{C} = (C, \boxplus, \perp, \circ, |\cdot|, \kappa)$ (standard model $C = \mathbb{R}^2$): superposition, rotation, amplitude, and a commitment cosine $\kappa \in [-1, 1]$ from which the balanced trit is *derived* with deadband $\delta = \frac{1}{2}$: $\tau = +1$ if $\kappa > \delta$, -1 if $\kappa < -\delta$, else 0; boundary openness $\varkappa = \max(0, 1 - |\kappa|/\delta)$ is maximal at the crossing. Fields of phase-cells in bounded modules evolve by continuous *flow* (phase turns toward the afferent superposition) and discrete guarded *commit* (quantize, barrier, belief step, energy guard, latch), with *nest* exchanging child coherence upward and context downward. The finite $n=6$ commit spectrum—729 words, 267 admissible, 51 localized (Motzkin), 20 saturated, 5 Catalan—is machine-checked and grounds GIST’s release tiers. The dyadic/triadic bridge ($2^6=4^3=64$) supplies GIST’s address system.

GIST rides CDC *subordinately*: its declaration files (`gist.cdc`, laws, lattice, pyramid, counterfactuals, gate, ladder, learning, neural, the training net, and the `.cdc` datasets and learned weights) are executed and checked by CDC’s own C runtime and Python bootloader. The kernel contract asserts `host-debt <= 0` and the verification gate confirms 83/83 expectations with 70 native witnesses. GIST is, to our knowledge, the first CDC-first system: the calculus describing a mechanism which the calculus’s own bridge then executes, verifies, and (below) learns.

3 The do-operator without a new primitive

A CDC channel graph *is* a structural causal model, so Pearl’s surgery [2] decomposes into two native acts: *latch* the intervened cell at its pole and *sever* its inbound channels. The counterfactual twin is a second field sharing initial conditions. Four runtime-checked programs exhibit the ladder of causation natively:

- **Necessity** (`gist_cf.cdc`): the factual field commits the conclusion $+$; under $do(r_0 := -)$ the twin dissolves it to the crossing— r_0 is load-bearing.
- **Sufficiency-restoration**: a contested conclusion (word $+ - 0$) is restored to $+$ by latching the opposing root.
- **Backdoor severance**: a confounder co-moves x and y (word $+++$); $do(x := -)$ with the backdoor severed yields only $+ - 0$ —the association overstates the causal effect.
- **Frontdoor mediation**: with the confounder still supporting y , $do(m := -)$ on the mediator alone dissolves the conclusion, isolating the mediated path.

Every expected phase and committed word in these files is checked by the C runtime to 10^{-5} . A companion reference implementation carries the distributional layer (d-separation, do-calculus rules 1–3 as executable rewrites, backdoor/frontdoor identification with honest refusal and positivity attribution, exact abduction/PN/PS over the enumerable ternary exogenous space) with

Table 1: Held-out results, 64-slot lattice (109 parameters; numpy on one CPU; 46 epochs chosen by early stopping). Majority baselines: slot 0.661, global 0.532.

	slot acc.	slot macro-F1	global verdict acc.
calculus init (untrained)	0.785	0.644	0.540
trained	0.984	0.980	0.868

16/16 suites passing, including exact agreement between identified estimands and surgical ground truth across random ternary SCMs.

4 The hybrid: the network the mechanism already is

Read as dynamics, GIST’s forward pass is a neural architecture in the precise, modern sense:

- **Recurrent nonlinear state space.** The flow step $\Delta\theta_i = g \operatorname{Im}(A_i e^{-i\theta_i})$ (Kuramoto turn [3]) is a recurrence whose coupling is state-dependent—the selectivity that distinguishes modern SSMS [4, 5] arises here from phase geometry itself.
- **Convolutional filtration.** Two weight-shared graph convolutions: the corefold pyramid inside every slot (six edge-type filters), and a Hamming-1 lattice convolution on the B -cube (one $\operatorname{SO}(2)$ filter per cube direction). Channels with weight w and angle α are exactly the blocks $w \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$.
- **Gating.** A learnable openness gate $\sigma(\beta(|\kappa| - \delta))$ scales each cell’s outbound influence: decisive cells transmit, crossings attenuate—the calculus’s aperture as a trainable gate.
- **Balanced-ternary bottleneck.** The commit quantizer with a straight-through gradient; activations end as $\{-, 0, +\}$ words behind the same deadband the mechanism uses (cf. ternary-weight LLMs [6], here as *semantic* quantization).
- **Ladder pooling.** B hypercube contractions with a shared alignment rotation per round pool 2^B slot singulars to one global state.

Native supervision. Training data is labeled by the mechanism itself: scenes compile to `.cdc` batch programs (evidence mass folded into channel weights, matching the runtime’s phase-only coupling), the C runtime commits them, and the committed words are the labels—negatives via the mirror rule (the barrier is assertion-leading, so “no” is a gated “yes” on the mirrored field).

Initialize at the calculus, train to anticipate. The provably-equivalent GIST-NN point (exact corefold weights, neutral angles, unit gain) is the initialization; training then learns to anticipate the *converged* native reduction from $K=4$ recurrent steps.

The ablations localize the contributions: slot-level accuracy is carried by the pyramid recurrence (removing lattice coupling even helps locally), while the *global verdict*—the judgment GIST exists to gate—is best served by the full hybrid with calculus initialization; random initialization and ablated gating cost 9 and 9.5 points of global accuracy respectively.

Table 2: Ablations (8-slot lattice, same protocol).

variant	slot acc.	global acc.
full hybrid, calculus init	0.951	0.840
random init	0.968	0.750
no lattice convolution	0.987	0.775
no openness gating	0.984	0.745
no ternary bottleneck	0.953	0.825

Deployment closes the loop. A parity-mode variant (gating off, amplitude-blind, matching the C runtime’s flow exactly) trains to 0.985 slot accuracy and exports its weights as a `.cdc` program; `cdc_native_runtime infer` then reproduces the trained model’s singular conclusion on *all* 27 ternary input words (27/27). The same runtime that verifies the mechanism runs the trained model: `./run.sh '+0-'`.

Training is native. The runtime extension executes CDC’s declared plasticity: channels marked `plastic` follow phase correlation through flow ($w += r(\cos(\theta_{src} - \theta_{dst}) - w) d$), witnessed by `gist_learn.cdc` (a weak channel learns an alignment, $0.10 \rightarrow 0.76$ in two steps, to 10^{-5}). On top of it the runtime’s `train` command implements clamped-teacher learning with no host code at all: the conclusion cell is latched at the label pole (a `do()`-style clamp) while plasticity adapts the weights through ordinary flow. Datasets are `.cdc` files whose labels the runtime’s own converged reduction computed; learned weights are written back as `.cdc`. On the anticipation task (two flow layers against a forty-step converged oracle), native training lifts held-out accuracy from 0.813 to **0.933** (400 training scenes, 10 epochs, deterministic). The shipped repository trains, evaluates, and infers entirely through the calculus and its C bridge; the gradient-based hybrid results above are construction-tier reference experiments whose harness deliberately stays out of the repository.

5 Adapters and integration surfaces

The reference tier exposes the mechanism to hosts: a one-call ternary verdict bridge (`yes/no/maybe` = the carrier, with “maybe” returning the asks that would settle it, and “no” derived Pearl-symmetrically via the mirrored session), an MCP server, an architecture-agnostic tool-call spec, and a reasoning-loop stitcher that feeds a model’s chain of thought through the lattice and injects the live field state back into context. The reference engine is replay-verifiable end-to-end (hash-chained ledgers, Merkle-rooted reductions, HKDF/HMAC decision stamps) and carries a 100-test suite plus a 36-witness contract of its own. A separate execution plan documents training the discipline into open-weight LLMs with the native runtime as the (near-free) RL reward function.

6 Reproducibility and limitations

Everything reported here reruns from the repository: `native/verify_gist.sh` (the whole native gate), `torchcdc/train.py` (datasets, training, ablations, export, parity; ~ 6 minutes on one

CPU), `torchcdc/test_torchcdc.py`, and `./run.sh`. Determinism is by construction (hash-derived streams; no wall clock in semantics). Limitations: the hybrid is small by design (the point is the isomorphism and the closed loop, not scale); the native runtime’s flow is phase-only (amplitude semantics are folded into weights at compilation, documented); identification tiers beyond backdoor/frontdoor (Shpitser–Pearl ID) remain reference-tier work; and the LLM translation is a designed protocol, not yet an executed run.

References

- [1] E. Ellis. *BiDi Coherence-Delta Calculus* (v0.2.4). <https://github.com/ETEllis/bidi-coherence-delta-calculus>, 2026.
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